Merits

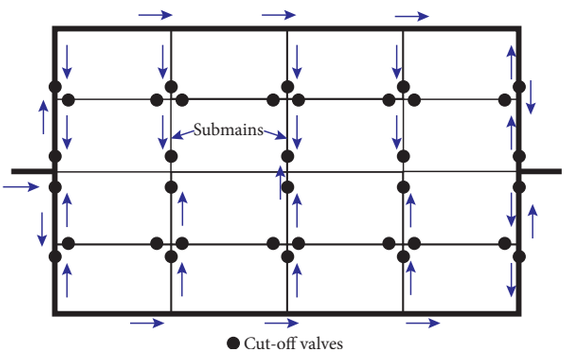
i. There is free circulation of water. No possibility of pollution due to stagnation.
ii. In case of repairs, only a very small portion of the distribution area will be affected.
iii. Water reaches all the points with minimum loss of head.

Demerits

i. Two or more valves are to be installed at every cross junction.
ii. Laying water pipes is costly.
iii. The design is difficult and costlier.

3. Circular or Ring system

In this system shown in fig, the entire distribution area is divided into square, rectangular or circular blocks. The water mains are laid on the periphery of these blocks. The submains, branches, etc., are laid along the inner roads and streets. This system is most suitable for cities having well planned roads and streets.



Merits

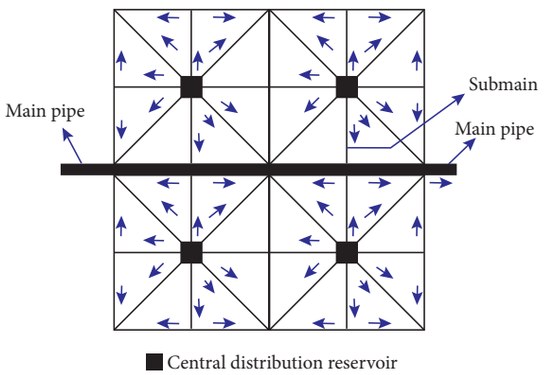
i. This system has the advantages of the previous two systems.
ii. In this systems, every point receive its supply from two directions.
iii. Design of this system is easy.

Demerits

This system has the same disadvantages as the grid-iron system.

4. Radial system

This system of layout shown in fig., is just the reverse of the ring system. The entire distribution area is divided into distributed zones. There is a central distribution reservoir in each zone. The supply pipes are laid radially away from the reservoir towards the periphery. The water from the mains flow into these reservoirs. From there it is supplied to the respective zones.



Merits

i. Provides good service quickly.
ii. Design calculations for the sizes of the pipes are simple.
iii. Best suited where the roads are laid radially.

Demerits

Requires a distribution reservoir in each zone.

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